

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. **Name of the Academic Unit:** Department of Bioscience and Biotechnology

2. **Subject Name:** Cell and Molecular Biology **L-T-P:** 3-1-0 **Credits:** 4

3. **Pre-requisites:** Science of Living Systems (BT10001)

4. **Syllabus and reference books:**

Syllabus:

Cell Biology

1. Introduction to Cell Biology: Cell Theory; Origin and evolution of cells; Prokaryotic vs eukaryotic cells; Various types of microscopes used in cell biology.
2. Membrane structure: Fluid Mosaic model; Membrane lipids, proteins; carbohydrates; Membrane compartmentalization.
3. Membrane Function: Differential permeability; Active and passive transport; Electrical properties of membrane.
4. Cellular organelles: Nucleus, ER, Mitochondria and Chloroplast; Protein sorting and Protein transport; Vesicular Transport; Lysosomal transport.
5. Cytoskeletal elements: Structure and function of microfilaments; microtubules; Cilia and Flagella; Intermediate filaments.
6. Chromatin composition and dynamics: Hierarchical packaging of DNA into chromosomes; Euchromatin and heterochromatin; Chromatin remodelling; Overview of epigenome.
7. Cell division and cell cycle: Mitosis, meiosis and different phases of cell cycle.
8. Cell Signalling: Autocrine, paracrine and endocrine cell signalling; Intracellular signalling proteins and effectors; Signalling through GPCR and enzyme coupled receptor.

Molecular Biology

9. Introduction to Molecular Biology: Nucleic acid structure; Chemical, physical, thermal and spectroscopic properties of nucleic acids; DNA supercoiling; and genome complexity.
10. DNA replication in prokaryotes: Origin, direction and model of DNA replication; Enzymes and proteins involved in DNA replication; DNA replication in E. coli, bacteriophage and adenovirus, and control of replication.
11. DNA replication in eukaryotes: Origin of replication; Enzymes and proteins involved; DNA replication in yeast and its control; Telomere and telomerase.
12. Transcription in prokaryotes: Promoters; RNA polymerase (structure and subunits), Promoter recognition, initiation, elongation and termination; Operon concept; Regulation of Lac operon and Trp operon.
13. Eukaryotic RNA polymerases; Eukaryotic transcription initiation by RNA polymerase I, II and III; 5' capping and 3' polyadenylation; Transcription factors and regulation of

eukaryotic transcription.

14. Eukaryotic mRNA splicing; Self splicing introns; Alternative RNA splicing; Processing of rRNA and tRNA.

15. Translation in prokaryotes and eukaryotes: Genetic code, Wobble base pairing; Ribosomes, tRNA structure, attachment of amino acid to tRNA: Translation initiation, elongation and termination; Post-translational modification of proteins. Concept of making recombinant DNA and its application.

Reference Books:

1) Cell Biology by Karp, Iwasa and Marshal, (2019), 9th edition, Wiley Publisher.

2) Molecular Biology by Clark, Pazdernik and McGehee, (2018), 3rd edition, Academic Cell Publisher

3) Molecular and Cell Biology by Lodish et al., (2021), 9th edition, W.H. Freeman and Co publisher.

4) Molecular Biology of the gene by Watson, (2017), 7th edition, Perason publisher

5) Lewin's Genes XII, by Krebs, Goldstein and Kilptrick, (2018), Jones and Bartlett Publisher.

6) Essential Cell Biology, (2019), 5th Edition, W. W. Norton & Company

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Introduction to Cell Biology: Cell Theory; Origin and evolution of cells; Prokaryotic vs eukaryotic cells; Various types of microscopes used in cell biology.	2
2.	Membrane structure: Fluid Mosaic model; Membrane lipids, proteins; carbohydrates; Membrane compartmentalization.	2
3.	Membrane Function: Differential permeability; Active and passive transport; Electrical properties of membrane.	2
4.	Cellular organelles: Nucleus, ER, Mitochondria and Chloroplast; Protein sorting and Protein transport; Vesicular Transport; Lysosomal transport.	3
5.	Cytoskeletal elements: Structure and function of microfilaments; microtubules; Cilia and Flagella; Intermediate filaments.	2
6.	Chromatin composition and dynamics: Hierarchical packaging of DNA into chromosomes; Euchromatin and heterochromatin; Chromatin remodelling; Overview of epigenome.	2
7.	Cell division and cell cycle: Mitosis, meiosis and different phases of cell cycle.	2
8.	Cell Signalling: Autocrine, paracrine and endocrine cell signalling; Intracellular signalling proteins and effectors; Signalling through GPCR and enzyme coupled receptor	3

9	Introduction to Molecular Biology: Nucleic acid structure; Chemical, physical, thermal and spectroscopic properties of nucleic acids; DNA supercoiling; and genome complexity	2
10	DNA replication in prokaryotes: Origin, direction and model of DNA replication; Enzymes and proteins involved in DNA replication; DNA replication in E. coli, bacteriophage and adenovirus, and control of replication.	3
11.	DNA replication in eukaryotes: Origin of replication; Enzymes and proteins involved; DNA replication in yeast and its control; Telomere and telomerase.	2
12.	Transcription in prokaryotes: Promoters; RNA polymerase (structure and subunits), Promoter recognition, initiation, elongation and termination; Operon concept; Regulation of Lac operon and Trp operon	3
13.	Eukaryotic RNA polymerases; Eukaryotic transcription initiation by RNA polymerase I, II and III; 5' capping and 3' polyadenylation; Transcription factors and regulation of eukaryotic transcription	3
14.	Eukaryotic mRNA splicing; Self splicing introns; Alternative RNA splicing; Processing of rRNA and tRNA.	2
15.	Translation in prokaryotes and eukaryotes: Genetic code, Wobble base pairing; Ribosomes, tRNA structure, attachment of amino acid to tRNA: Translation initiation, elongation and termination; Post-translational modification of proteins. Concept of making recombinant DNA and its application.	3
16.	Tutorial	12
Total number of hours		